HUD HACKATHON · RL FOR HARDWARE

FizzVolt

We turned the EDA toolchain into a **reward function.**

Reinforcement learning for verifiable, manufacturable circuit design.

WHY IT'S HARD

One chip. Ten exact connections. *Every time*.

FIG 7-1 · TYPICAL APPLICATION — STEP-DOWN REGULATOR

1. `C_IN` ≥ 10 µF within 2 mm of VIN→GND
2. Bootstrap cap `C_BST` 0.1 µF, SW→BST
3. Divider sets `V_OUT = 0.8·(1+R1/R2)`
4. Catch Schottky SW→GND, **cathode to SW**
5. Inductor on SW; `C_OUT` on VOUT→GND
6. EN pull-up to VIN; keep FB trace short

…and that's one regulator. Now the sensor, the MCU, the USB-C front end…

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INTERDEPENDENT PARTS / IC

1

WRONG PART = 1 RE-SPIN

Days–weeks lost.

Blocked team. Delayed launch.

WHY AGENTS FAIL TODAY

LLMs write schematics that look right — and don’t build.

- ✕ `D1.1 / D1.2` — LED pins are actually **A / K**
- ✕ USB-C `CC2` floating — **won’t enumerate**
- ✕ MCU `VDD` with **0 decoupling caps**
- ✕ Buck stage — **no inductor, no catch diode**
- ✕ Reasoned for 12 KB of tokens, **emitted no code**

Real failures logged from a strong base model.

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RAW BUCK ATTEMPTS THAT BUILT

Plausible ≠ buildable.

No verifiable signal → nothing to learn from.

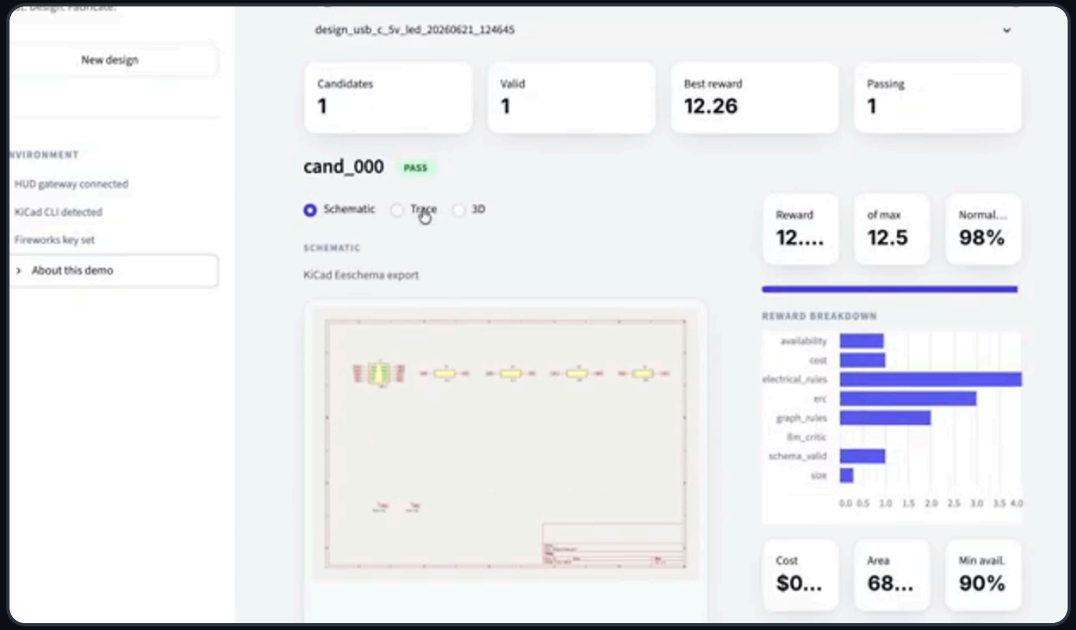
THE SOLUTION

Make the EDA toolchain the source of truth.

Prompt → **Circuit DSL** → **Real KiCad**

ERC + validators → **Reward 0–1**

- **Sandboxed DSL** compiles *deterministically* to real KiCad.
- **ERC + validators**: correctness, required subcircuits, cost · area · availability · solderability.
- Every check → one **0–1 reward**.
- A reproducible **RL gym — 31 tasks**, LED → switching regulators.



Graded against real KiCad — ERC 0 errors · reward 12.26 / 12.5.

THE LEARNING LOOP

GRPO turns verification into improvement.

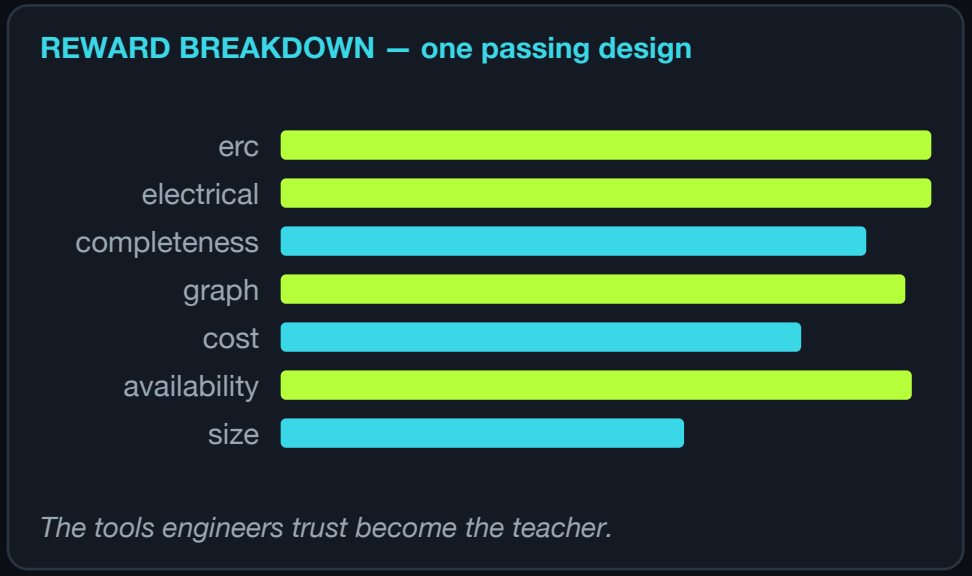
- Sample many designs/task; **reward only those that pass** — no labels, no SPICE.
- A **rich** reward: ERC · electrical · **completeness** · cost · size · availability.
- **Within-group variance is the gradient** — sometimes right, sometimes wrong, so it learns.
- Trained on **HUD** (GRPO · Qwen3-8B / Tinker), graded by **real KiCad**.

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REWARD DIMENSIONS

98%

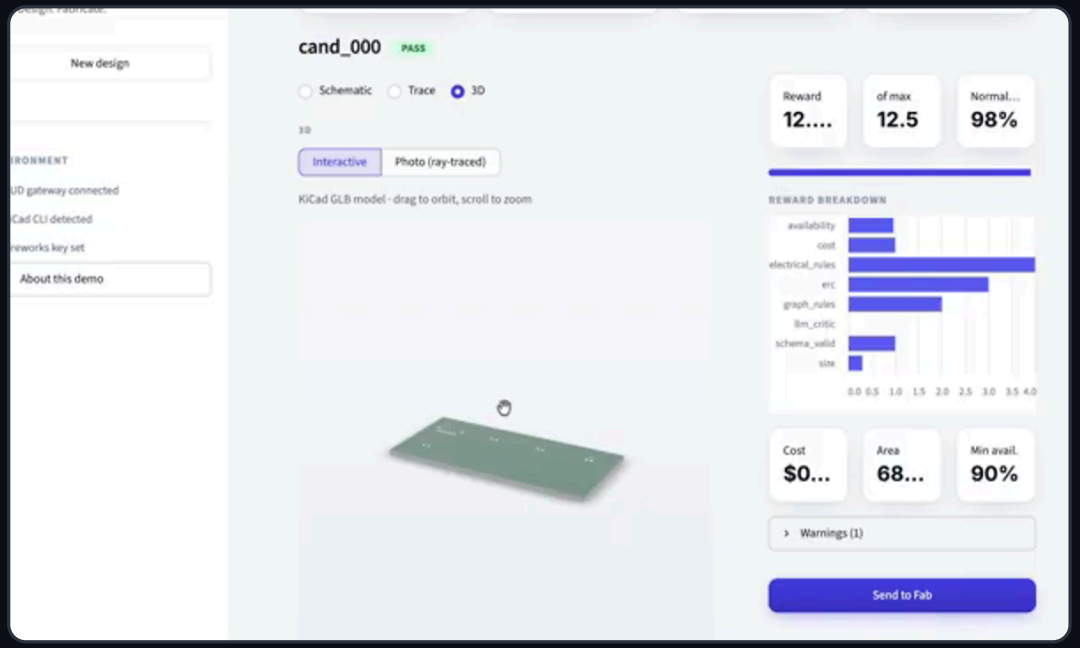
OF MAX ON PASSES



The tools engineers trust become the teacher.

THE IMPACT

From a sentence to a manufacturable board.



Schematic → Trace → 3D → **Send to Fab**, graded end-to-end.

- Attacks a **massive bottleneck**: repetitive datasheet work → **validated automation**.
- Trusted tools become **automated feedback** → less review, **fewer re-spins**.
- Compresses **idea → validation → manufacturable hardware**.
- Next: **PCB layout · placement · routing**.

Don’t just look right — build.